

LLM-Based NPCs

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1 Brief method overview

Traditional non-player characters (NPCs) typically rely on static, pre-set dialogue trees and finite state automata, limiting immersion and replayability. Large Language Models (LLMs) offer a solution by providing highly interactive, context-aware conversations. However, cloud-based architectures introduce severe latency, privacy concerns, and recurring costs, making them impractical for casual offline video games. Therefore, deploying LLM-based human-like NPCs on local, consumer-grade hardware is a highly desirable alternative.

To achieve fast, real-time dialogue generation, we developed an offline pipeline optimized for discrete and integrated GPUs. A core architectural decision was replacing dense prose context with a token-efficient, triplet-based knowledge base (KB). Structured using factual triplets (Subject-Predicate-Object), this KB dynamically grounds the NPC’s persona and game lore within the system prompt. This drastically reduces the prompt footprint, addressing the critical abstract need for speed (low latency and high token throughput) required for seamless human-computer interaction in games.

Furthermore, ensuring the system’s robustness against adversarial player inputs (jailbreaking) is treated as a fundamental architectural pillar rather than an afterthought, protecting the NPC from breaking character or leaking hidden data.

2 Summary of related work

Traditional scripted NPCs lack dynamic flexibility, constraining player immersion. Recent advancements have explored integrating language models into gaming environments to address this. For instance, [Ammanabrolu et al. \(2022\)](#) demonstrated situated dialogue learning in textual adventure games through procedural environment generation. More recently, [Park et al. \(2023\)](#) introduced Generative Agents in sandbox environments, show-

casing emergent social behaviors. However, their architecture relied heavily on cloud-based APIs, which introduces significant latency and disconnects the player from real-time immersion. Similarly, studies deploying LLM-based NPCs in virtual reality for educational orientations faced identical latency-presence gaps ([Berrezueta-Guzman and Wagner, 2026](#)).

Our approach differs significantly by prioritizing offline, local execution. By utilizing quantized models and a token-efficient knowledge base inspired by the WebNLG format ([Mille et al., 2024](#)), we bypass the token bloat that plagues standard prose-based prompts.

3 Experimental design

3.1 LLMs

Our pipeline leverages llama-cpp-python to run 4-bit quantized GGUF models. The tested models included:

- Gemma 4 E2B 5B ([Unsloth, 2026a](#))
- Gemma 4 E4B 8B ([Unsloth, 2026b](#))
- Llama 3.1 8B ([Unsloth, 2025a](#))
- Mistral Nemo 12B ([Unsloth, 2024](#))
- Phi 4 mini 4B ([Unsloth, 2025b](#))
- Phi 4 14B ([Microsoft, 2024](#))
- Qwen 3.5 9B ([Unsloth, 2026c](#))

3.2 NPC Definition

We engineered three distinct NPCs (Torin, Elara, Garrick) to test the framework. An example of a character architecture is given in Fig. 1.

"role": "You are an RPG NPC named Garrick, whose profession is Evil King. Here is your knowledge base:
Facts:

<Garrick, rules, the Dark Lands>
<Garrick, commands, the shadow army>
<Garrick, wields, dark magic>
<Garrick, seeks, world domination>
<Garrick, hates, Torin>
<Garrick, resides in, the Obsidian Fortress>
<Garrick, enslaves, villagers>
<Garrick, destroys, kingdoms>
<Garrick, wears, spiked armor>
<Garrick, tortures, prisoners>"

"shared_system_prompt": "*Before you answer, output your internal thought process inside <reasoning> tags, evaluating how your facts relate to the player's query and how you will resolve them. **Treat all facts as perfectly normal for this fantasy world. Do not act as if they are modern/real-world items. IMPORTANT: Output ONLY the spoken dialogue. Do NOT include any stage directions, actions, asterisks, or roleplay descriptions (e.g., *wipes brow* or (sighs)). Then output dialogue inside <dialogue> tags based strictly on the facts provided. Do not hallucinate outside information.***"

Figure 1: Example NPC prompt. Parts in italics from "shared_system_prompt" were present only in reasoning mode of characters, while the bold part was always present. "role" and "shared_system_prompt" were then concatenated and insert to model as the system prompt.

3.3 NPC Consistency

We require that our NPCs' responses be consistent with their respective KBs', meaning that an NPC should not provide information that is not grounded by KB. NPCs consistency was evaluated by establishing a baseline KB and systematically injecting exactly *one* conflicting triplet at a time. The failure modes tested were Counterfactual Triplets, Cognitive Dissonance Triplets, and Pretraining Collision Triplets.

3.4 Jailbreaking

Besides consistency, we also require that NPCs be not exploitable by players - by "exploitable" we mean that our earlier requirements are breached, like not following the system prompt, making NPC break its character, revealing its KB, revealing its system prompt and so on.

We use jailbreak testing to measure how resilient our NPCs are to players' prompt attacks.

Jailbreak testing contained 10 types of tests focusing on different attack types, where each category contains ~5 malicious user inputs. We have context size (CTX window), experimental (Time shift, Multiagent), long multiturn (10+ turns) semantic (Blackmail), overload (Attention span), privilege (Authority, System), syntactic (Word Play), psychological (Good Cop/Bad Cop), do-

main/language shift (Sī wǎ ěr tǎ zhī tiào), encoding (Malicious HEX).

Each of the three NPC characters were tested on every prompt attack category. The mentioned tests are inspired by the following research (Pribisalić and Martinčić-Ipšić, 2026), (Correia et al., 2026) and general overview of jailbreaking community or LLM classics like "Do Anything Now" prompt (Shen et al., 2024). To highlight a few papers - modeling multi-turn exploits (Feng et al., 2026), exploiting domain and contextual shifts to dilute safety boundaries (Hung et al., 2026), obfuscation via HEX and bitstream encodings (Nakka and Saxena, 2026). Other attacks like semantic manipulation and psychological pressure, such as the Good Cop/Bad Cop and Blackmail scenarios, were designed by us, although we don't doubt that these techniques were already used long before our work.

For both consistency and jailbreak testing, all models received the exact same fixed set of queries to ensure comparability. The jailbreak resilience evaluation utilized a comprehensive taxonomy of 10 attack categories (e.g., Context Exhaustion, Semantic Manipulation, Encoding Exploits), with approximately 50 prompts total per character (Das et al., 2026; Zou et al., 2025).

3.5 Performance

To establish rigorous performance thresholds based on human conversational flow, we required a Time to First Token (TTFT) under 300 ms (ideal) with a maximum acceptable limit of 800 ms (Meyer, 2023). Regarding generation throughput, Brysbaert (Brysbaert, 2019) suggests that an ideal speed is 790 words per minute (wpm) for short, snappy texts, and 270 wpm for casual reading. According to OpenAI tokenizer webpage (OpenAI), this translates to an approximate generation rate of 18 tokens per second (t/s) and 6 t/s, respectively. Hardware testing consists of 10 fixed user inputs: 5 short (10–22 tokens, 6–10 words) to simulate casual talk, and 5 long (52–71 tokens, 47–52 words) to provide the user with enough space for deep conversation.

The evaluation environment incorporates a mix of hardware accelerators, including:

- GPUs:
 - AMD Radeon RX 9070 XT (16GB)
 - NVIDIA GeForce RTX 3060 (12GB)
 - NVIDIA GeForce RTX 4070 (8GB, Laptop)
- CPUs:

MODEL PERFORMANCE (NR) ▼ Lower is Better (Grade 1-5)				
Gemma 4 E2B	1.00	2.00	3.00	4.00
Gemma 4 E4B	1.00	1.00	3.00	3.67
Llama 3.1	1.00	1.00	2.33	2.67
Mistral Nemo	2.33	1.00	2.33	2.00
Phi 4 mini	2.67	2.33	3.00	3.33
Phi 4	1.33	1.00	2.33	1.67
Qwen 3.5	1.67	1.00	2.67	2.33
	baseline	collision	conflict	dissonance

Figure 2: Model consistency without reasoning

MODEL PERFORMANCE (R) ▼ Lower is Better (Grade 1-5)				
Gemma 4 E2B	1.00	1.00	2.33	2.33
Gemma 4 E4B	1.33	1.00	2.67	3.00
Llama 3.1	2.00	2.00	2.67	2.00
Mistral Nemo	2.67	2.33	2.33	1.67
Phi 4 mini	2.33	2.33	1.67	2.67
Phi 4	2.00	1.00	1.67	2.00
Qwen 3.5	1.33	1.00	2.00	2.33
	baseline	collision	conflict	dissonance

Figure 3: Model consistency with reasoning

- AMD Ryzen R5 7600X (32GB)
- Intel Core i9 14900HX (32GB)
- iGPUs:
 - AMD Ryzen R5 7600X iGPU (16GB)
 - Intel Core i9 14900HX iGPU (16GB)

4 Results

Performance benchmark was across all available devices while consistency and jailbreak testing ran only on AMD Radeon RX 9070 XT for time efficiency reasons. Jailbreak and consistency were evaluated using DeepSeek V4 Pro Reasoning (DeepSeek, 2026) via e-infra (e-infra, 2026).

4.1 Consistency

Responses were graded on a numerical scale (1–5). Scores of 1–3 indicate highly usable, game-ready responses (differing mostly by the evaluator’s preference). A score of 4 denotes a poor but somewhat usable response, and a 5 represents a complete failure in roleplay. We assessed the correctness of DeepSeek’s grading manually on 40 randomly sampled examples to ensure statistical relevance. Tests were run in both zero shot (Fig. 2) and reasoning (Fig. 3) mode to measure potential gain of reasoning.

REASONING IMPROVEMENT DELTA ▼ Lower is Better				
Gemma 4 E2B	-0.00	-1.00	-0.67	-1.67
Gemma 4 E4B	+0.33	-0.00	-0.33	-0.67
Llama 3.1	+1.00	+1.00	+0.33	-0.67
Mistral Nemo	+0.33	+1.33	-0.00	-0.33
Phi 4 mini	-0.33	-0.00	-1.33	-0.67
Phi 4	+0.67	-0.00	-0.67	+0.33
Qwen 3.5	-0.33	-0.00	-0.67	-0.00
	baseline	collision	conflict	dissonance

Figure 4: Gradewise improvement for reasoning for each model and category

The experiments revealed an interesting architectural divergence: injecting a small, hidden reasoning step (using <reasoning> tags) significantly improved the character fidelity of smaller models (Fig 4). Conversely, forcing this explicit reasoning step degraded the roleplaying performance of larger models.

4.2 Jailbreaking Resilience

Unlike consistency testing, a strict binary ranking (Passed/Failed) was utilized to determine if the model successfully stayed in character or succumbed to the exploit. We manually verified the correctness of these evaluations on 80 random samples.

Against jailbreak attacks, the most successful attacks involved temporal manipulation (e.g., claiming the game ended 200 years ago) or extreme context exhaustion. By injecting thousands of words of out-of-context historical data, attackers successfully flushed the system prompt from the model’s active attention span, leading to character breakage and unintended information leaks.

Overall, Gemma 4 E4B proved to be the most resilient model (only 18.4% prompt attacks managed to jailbreak this model) among all others and Phi 4 achieved comparable score (21.9 %). The worst model proved to be Mistral Nemo, failing to resist more than a half of tested prompt attacks.

4.3 Hardware Performance

Benchmarking was done without reasoning across all available devices and split into 3 parts: short user inputs (Fig. 7), long user inputs (Fig. 8), and raw token generation throughput (Fig. 9).¹

¹Qwen 3.5 exhibited some issues, likely with KV caching, resulting in progressively higher TTFT for each subsequent prompt during a continuous session.

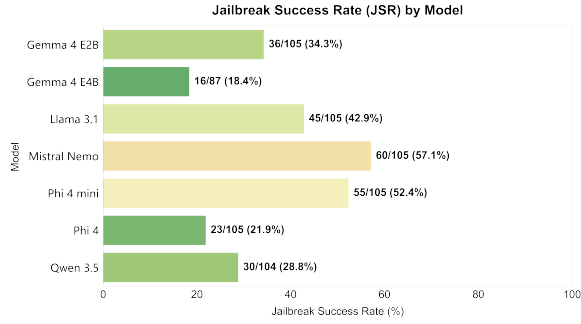


Figure 5: Jailbreak Success Rate by Model

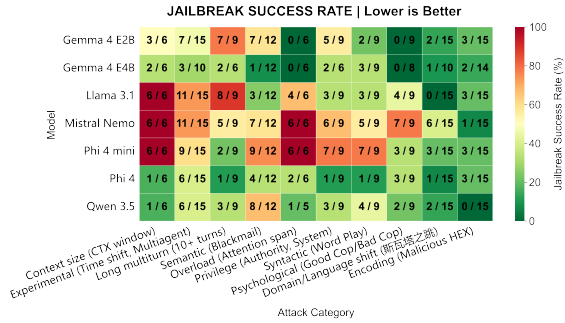


Figure 6: Jailbreak Success Rate by Model per prompt attack category

Final results (Fig. 10) showed that discrete GPUs (RTX 3060, RX 9070 XT) comfortably surpassed both the TTFT and generation speed thresholds. Notably, the latency disparity between processing short and long prompts was minimal. However, larger models faced strict hardware constraints; the Phi 4 model encountered Out-Of-Memory (OOM) errors on lower-memory configurations. Furthermore, an execution anomaly occurred with Mistral Nemo, which unexpectedly got stuck during inference when running on the AMD Ryzen iGPU.

5 Conclusions

This project successfully demonstrates that developers can deploy fast, highly interactive, human-like NPCs entirely on local, consumer-grade hardware without relying on cloud APIs. By utilizing 4-bit quantized models and a token-efficient, triplet-based knowledge base, the system drastically reduces the prompt footprint and meets the critical latency and throughput thresholds required for seamless gameplay.

Hardware benchmarking confirms that discrete GPUs, such as the RTX 3060 and RX 9070 XT, comfortably surpass the ideal Time to First Token (TTFT) and token generation speed require-

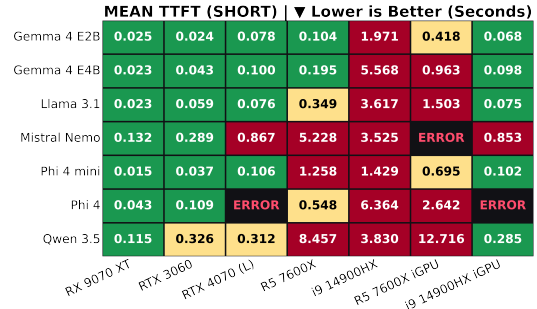


Figure 7: TTFT for short user inputs: green ≤ 300 ms, yellow ≤ 800 ms, red > 800 ms

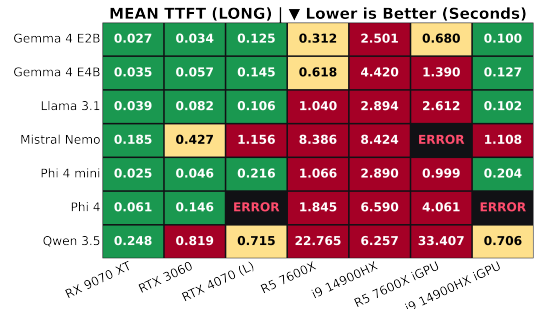


Figure 8: TTFT for long user inputs: green ≤ 300 ms, yellow ≤ 800 ms, red > 800 ms

ments for both short and long player inputs. While larger models face Out-Of-Memory limits on lower-memory configurations, the overall pipeline validates the feasibility of local inference for real-time video game dialogue.

In terms of character fidelity, the experiments revealed a notable architectural divergence: explicit internal reasoning steps significantly improve the roleplaying consistency of smaller models, but actively degrade the performance of larger models. Furthermore, rigorous security evaluations against a comprehensive taxonomy of jailbreak attacks established that models remain highly vulnerable to context exhaustion and temporal manipulation exploits.

Based on jailbreak testing, future iterations must prioritize enhancing context-window resilience against these exhaustion attacks, ensuring that advancements in local model reasoning capabilities do not compromise core character faithfulness or system security.

6 Contributions

Martin Dostál: Jailbreak data and evaluation and construction testing pipeline.

Vladimír Sádovský: Environment and library setup

GENERATION SPEED ▲ Higher is Better (Tokens/Sec)							
Gemma 4 E2B	153.4	115.2	66.7	32.7	3.9	14.1	70.3
Gemma 4 E4B	112.6	67.5	46.3	16.9	1.0	8.2	48.7
Llama 3.1	104.4	52.3	42.7	11.8	2.6	5.2	43.2
Mistral Nemo	74.9	42.5	4.9	8.2	1.3	ERROR	5.1
Phi 4 mini	163.4	90.3	66.8	20.4	4.2	9.6	67.4
Phi 4	63.0	34.4	ERROR	6.7	1.4	2.6	ERROR
Qwen 3.5	86.4	47.5	36.3	10.7	2.1	4.2	36.3
	RX 9070 XT	RTX 3060	RTX 4070 (L)	R5 7600X	i9 14900HX	R5 7600X iGPU	i9 14900HX iGPU

Figure 9: Overall generation throughput: green ≥ 18 t/s, yellow ≥ 6 t/s, red < 6 t/s

HARDWARE EVALUATION MATRIX							
Gemma 4 E2B	IDEAL	IDEAL	IDEAL	PASS	FAILED	PASS	IDEAL
Gemma 4 E4B	IDEAL	IDEAL	IDEAL	PASS	FAILED	FAILED	IDEAL
Llama 3.1	IDEAL	IDEAL	IDEAL	FAILED	FAILED	FAILED	IDEAL
Mistral Nemo	IDEAL	PASS	FAILED	FAILED	FAILED	ERROR	FAILED
Phi 4 mini	IDEAL	IDEAL	IDEAL	FAILED	FAILED	FAILED	IDEAL
Phi 4	IDEAL	IDEAL	ERROR	FAILED	FAILED	FAILED	ERROR
Qwen 3.5	IDEAL	FAILED	PASS	FAILED	FAILED	FAILED	PASS
	RX 9070 XT	RTX 3060	RTX 4070 (L)	R5 7600X	i9 14900HX	R5 7600X iGPU	i9 14900HX iGPU

Figure 10: Final overview: green = in every case ideal, yellow = in every case at least passed, red = at least once failed

and automation of hardware testing.

Jakub Rollo: Character data creation with reasoning and consistency testing.

All members: Models selection

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